

Question			Marking details	Marks available				Maths	Prac
				AO1	AO2	AO3	Total		
			1-2 marks Comprehensive account including reference to one of increase in temperature, Newton's laws of motion and increase in pressure or limited account of 2 areas. <i>There is a basic line of reasoning which is not coherent, largely irrelevant, supported by limited evidence and with very little structure.</i> 0 marks <i>No attempt made or no response worthy of credit.</i>						
	(b)	(i)	$\text{rms} = \sqrt{\frac{400^2 + 425^2 + 450^2 + 550^2 + 625^2}{5}} \quad (1)$ $= 497 \text{ [m s}^{-1}\text{]} \quad (1)$		2		2	2	
		(ii)	The expected rms is explained by: $PV = nRT \quad \text{so} \quad P = \frac{nRT}{V}$ $\rho = \text{mass}/V = \frac{n(M_r \times 10^{-3})}{V} \quad \text{use of both equations by substitution (1)}$ substitute these into: $P = \frac{1}{3}\rho\overline{c^2}$; $\sqrt{\overline{c^2}} = \sqrt{\frac{3P}{\rho}}$; $\sqrt{\overline{c^2}} = \sqrt{3 \frac{nRT}{V} \frac{V}{n(M_r \times 10^{-3})}} = \sqrt{\frac{3RT}{(M_r \times 10^{-3})}}$ $\sqrt{\overline{c^2}} = \sqrt{\frac{3(8.31)(293)}{(32 \times 10^{-3})}} = 478 \text{ [m s}^{-1}\text{]} \quad (1)$ Valid conclusion with data i.e. yes, the rms speed of the five molecules is slightly higher [about 4% above the expected rms of the gas] (1) ecf Alternative solution: Use $m\overline{c^2} = 3kT$ $\sqrt{\overline{c^2}} = \sqrt{\frac{3kT}{m}} = \sqrt{\frac{3(1.38 \times 10^{-23})(293)}{(32)(1.66 \times 10^{-27})}}$ with sensible substitution (1) = 478 [m s⁻¹] (1)		1	1	3	3	